Investigating Speech Production in Aphasia: Usage-Based Approach

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Usage-based theories propose that speakers rely on linguistic knowledge for real-time speech production. For individuals with aphasia (IWA), reliance on high-frequency, entrenched utterances may serve as a compensatory mechanism to enhance fluency [1]. Recent work demonstrated that high-frequency phrases resulted in significantly fewer and shorter pauses in aphasic speech [1]. In the present study, we replicated and extended this work using the AphasiaBank corpus to investigate whether lexical frequency and bigram collocation strength predict pause and word duration in naturalistic speech, and how these effects differ among speakers with Broca's aphasia, Conduction aphasia, and post-stroke Non-Aphasic controls.

Naturalistic speech samples from 34 participants (9 Broca, 13 Conduction, 12 Non-Aphasic) from the AphasiaBank corpus (cite) were analyzed, with a total of 24,537 words of speech (Broca: 5,541, Conduction: 8,430, Non-Aphasic: 17,199). Speech was manually segmented in Praat [2] to facilitate extraction of silent pause durations (inter-word silence > 250ms) and word durations. Durations larger than 3 SD above or below the mean for each participant were excluded from analysis. Log-transformed lexical and bigram frequency values were extracted from the Corpus of Contemporary American English (CoCA) [4]. Hypotheses were tested using Linear Mixed-Effects Models (LMMs) with random intercepts for participants to account for the nested data structure.

Results demonstrated that speakers with Broca's aphasia produced the highest-frequency words, consistent with compensatory reliance on entrenched lexical items when cognitive resources are limited. This group also generally showed the longest silent pauses, reflecting their effortful, non-fluent speech. Crucially, a significant main effect across all groups was observed: increasing collocation strength significantly predicted shorter silent pauses (β = -0.091, p < 2e-16) and shorter word durations (β = -0.120, p < 2e-16). This result confirms the usage-based theory that highly frequent multiword units facilitate fluency. However, the hypothesis that IWA would show the greatest frequency-based facilitation was not consistently supported. While speakers with Broca's aphasia showed the maximal reduction in silent pause duration for strong bigrams (steepest slope), recovered Non-Aphasic speakers also demonstrated strong sensitivity to collocation frequency. Furthermore, speakers with Conduction aphasia showed comparable or larger effects than Broca's aphasics, indicating that clinical fluency labels are poor predictors of fine-grained prosodic behavior.

Results provide some support for usage-based theories by demonstrating that frequency-based facilitation is a predictor of smoother speech in both aphasic and recovered non-aphasic speakers. Importantly, the observed variability in effect size across groups, with the greatest compensatory efficiency shown by Broca's speakers but not consistently the largest magnitude effect, indicates that a solely usage-based model is insufficient. Instead, fluency possibly

emerges from the interaction of ingrained linguistic knowledge, cognitive processing, and the neural constraints imposed by damage and subsequent network reorganization. This work goes beyond traditional Chomskyan rule-based deficit models by establishing that aphasic speech is modulated by experience-driven expectations, offering a path toward personalized, data-informed methods for enhancing fluency in rehabilitation.

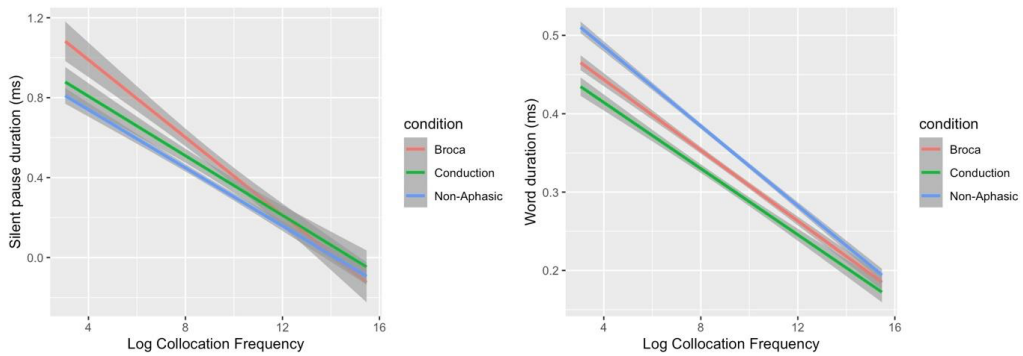

Figure 1. Left: Silent pause duration as a function of bigram frequency across groups. Broca's speakers exhibited the steepest negative slope, indicating the largest reduction in pauses with increasing predictability. Conduction and Non-aphasic speakers showed flatter slopes, consistent with less reliance on predictability during planning. Right: Scaled word duration (normalized for participant baseline) as a function of bigram frequency. Broca's speakers show the greatest reduction in scaled word duration for predictable bigrams, while Conduction speakers exhibit a flatter slope, reflecting reduced articulatory efficiency relative to predictability.

	silentpause			
Predictors	*Estimates*	*std. Error*	*t*	*p*
(Intercept)	1.37	0.11	12.27	**<0.001**
Bigram Frequency	-0.09	0.01	-16.72	**<0.001**
Broca's vs Conduction	-0.20	0.16	-1.23	0.219
Broca's vs Non-aphasic	-0.37	0.14	-2.70	**0.007**
Freq x Broca/Conduction	0.02	0.01	2.16	**0.031**
Freq x Broca/Non-aphasic	0.02	0.01	3.75	**<0.001**
Random Effects				
σ^2	1.60			
$\tau_{00\ participant}$	0.06			
$N_{participant}$	24			
Observations	24555			

Table 1. The model examines the interaction between Bigram Frequency (log-transformed) and Aphasia Condition on the duration of silent pauses, which indexes the cost of speech planning. The significant positive estimates for both interaction terms (Freq x Broca/Conduction; Freq x Broca/Non-aphasic) confirm that these groups have flatter (less negative) slopes than Broca's baseline, demonstrating Broca's group's maximal compensatory efficiency in reducing planning costs.

References

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